Hardware Design Checklist and Guidelines: Ti60 F225 FPGA, SPI Active Parallel (x4) Configuration

Power Pin Checklist

Check	Pin Name	Package Pins	Description
☐	All	All	My system uses the recommended number of decoupling capacitors. See guideline.
☐	All	All	I have used the power estimator to ensure that the system power requirements are met.
☐	All	All	My system follows the power-up sequence. See guideline.
☐	All	All	My system follows the connection requirements for unused resources and features. See guideline.
☐	VCC	E13 G7 G9 H6 H8 J7 J9 M3	The core power supply voltage is 0.95 V +/- 30 mV or 0.85 V +/- 30 mV.
☐	VCCAUX	D4 E11 L4 M11	The auxiliary power supply is 1.8 V +/- 50 mV.
☐	VCCA_xx	E5 G10 L10 L5	☐ The PLL analog power supply voltage is 0.95 V +/- 30 mV or 0.85 V +/- 30 mV. ☐ The PLL analog power supply has a separate filtering circuit.
☐	VCCIO33	B13 B4 N15 P4	The HVIO bank power supply is 1.8 V, 2.5 V, 3.0 V, or 3.3 V.
☐	VCCIOxx	C11 C7 D2 D6 D9 F14 F4 G12 J11 J4 L14 L2 N11 N5 N7 N9	☐ The HSIO bank power supply to HSIO configured as GPIO is 1.2 V, 1.35 V, 1.5 V, or 1.8 V. ☐ The HSIO bank power supply to HSIO configured as LVDS is 1.8 V. ☐ The HSIO bank power supply to HSIO configured as MIPI lane is 1.2 V. ☐ The HSIO bank power supply matches the external configuration interfaces. ☐ The system does not mix different I/O voltages in the same bank.
☐	VQPS_GND	G6	VQPS_GND must always stay low unless you are blowing the security fuses. See guideline.

Power Pin General Guidelines

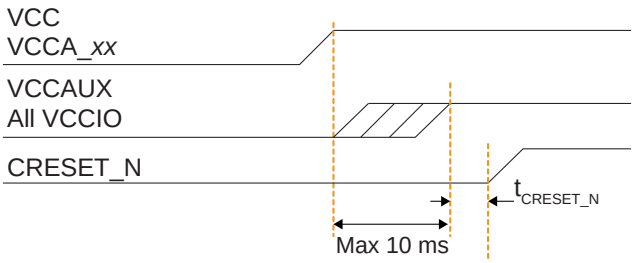
Recommended Decoupling Capacitors

When designing your board, Efinix recommends that you place the decoupling capacitors as close as possible to the power pins.

Power Supply	100 uF	47 uF	10 uF	4.7 uF	1 uF	0.1 uF	0.01 uF	1 nF
VCC	1	1	0	1	1	5	5	2
VCCA (1) (2)	0	0	0	0	0	2	2	1
VQPS	0	0	0	0	2	1	1	1
VCCAUX	0	0	0	1	0	1	1	1
VCCIO (1) (2)	0	1	1	1	1	1	1	1
VCCIO33 (1) (2)	0	0	0	1	1	1	0	0

Notes:
1. The number of recommended decoupling capacitors for each bank.
2. For merged banks, multiply the recommended number of decoupling capacitors by the number of merged banks. (e.g., if the merged bank is 4A_4B, there are 2 merged banks; multiply the number of recommended decoupling capacitors by 2.)

Power Up Sequence



1. Power-up VCC and VCCA_xx first.
2. When VCC and VCCA_xx are stable, power-up all VCCIO and VCCAUX. There is no specific timing delay between the VCCIO pins
   Important: Ensure the power ramp rate is within VCCIO/10 V/ms to 10 V/ms.
3. After all power supplies are stable, hold CRESET_N low for a duration of tCRESET_N before asserting CRESET_N from low to high to trigger active SPI programming (the FPGA loads the configuration data from an external flash device).

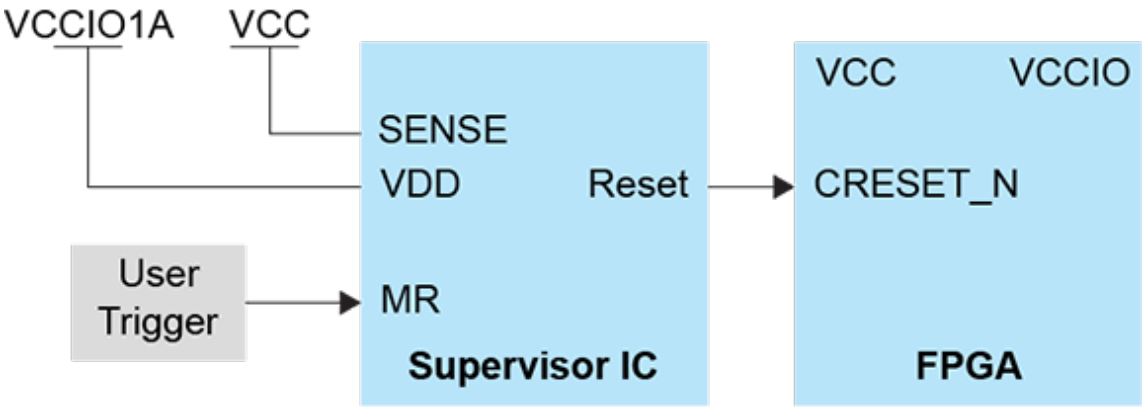
Holding CRESET_N Low

Use one of the following methods to hold the `CRESET_N` pin of the FPGA low after the power supplies are stable:

- Supervisor integrated circuit (IC)
- Microprocessor or microcontroller

Important: Do not drive a signal to any Titanium I/O pins before the Titanium FPGA is powered up. Most FPGAs have electrostatic discharge (ESD) circuits to protect the devices from ESD events. Driving the I/O pins before VCCIO will result in an in-rush current driving the I/O pins to a specific voltage through the ESD circuit to the VCCIO rail. Titanium FPGAs will remain in configuration mode after power-up if this unexpected voltage exists on `CRESET_N` due to the improper power-up sequence.

Supervisor IC Circuitry Example

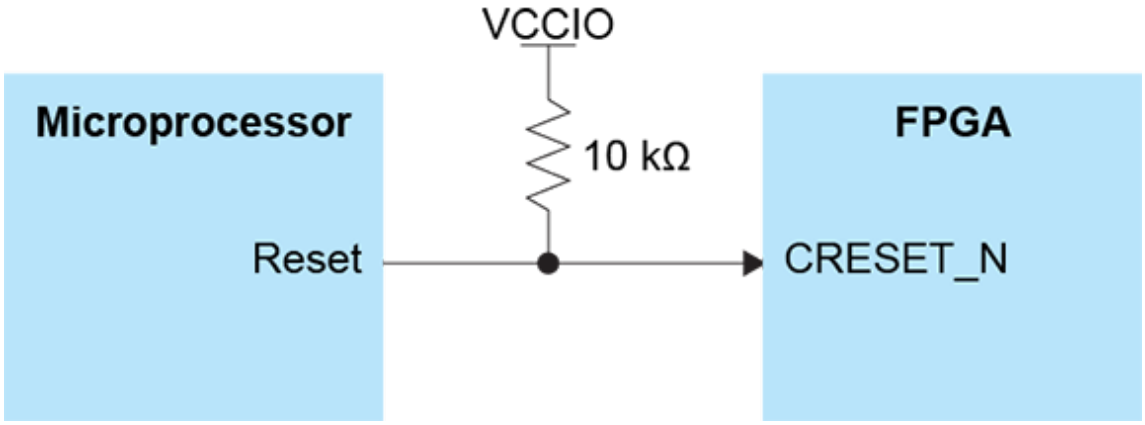


Assuming that the VCCIO1A is the last power supply to be stable in the system, the supervisor IC must hold the `CRESET_N` pin low for a duration of t_{RP} (reset timeout period) after the VCCIO1A reaches the stable threshold.

Ensure that the t_{RP} of the selected supervisor IC is more than the required t_{CRESET_N}. Refer to the supervisor IC vendor for the recommended operating circuitry.

Note: The user trigger (push-button, FTDI cable or module) must be connected to the MR pin of the supervisor IC.

Microprocessor or Microcontroller Circuitry Example



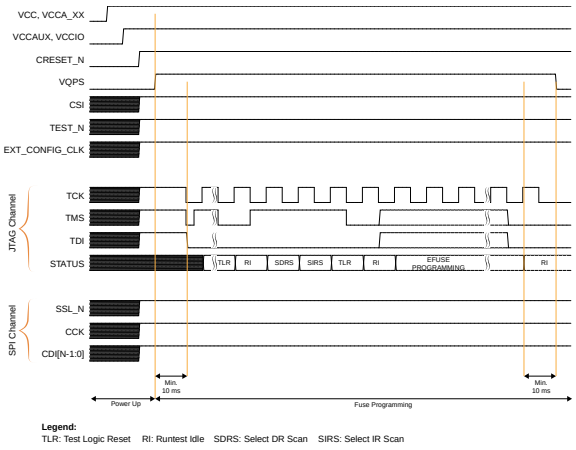
The microprocessor or microcontroller must hold the `CRESET_N` pin low more than the required t_{CRESET_N} duration.

Fuse Programming Requirements

Note: These requirements apply to F100S3F2 and F225 packages with the letter **S** in the lot number.
F100S3F2 and F225 packages that **do not** have the letter **S** in the lot number do not have these requirements.

To program the security fuses, follow these requirements:

- During fuse programming, avoid device configuration and other JTAG operations which are not related to fuse programming.
- Ramp up `VQPS` only after all the other power pins have ramped to their nominal voltages.
- After powering up `VQPS`, wait for a minimum of 10 ms before issuing JTAG instructions for fuse programming.
- Upon completion of the fuse programming through JTAG, wait for a minimum of 10 ms before powering down `VQPS`.
- If required, other power supplies can be powered down only after `VQPS` has been powered down below 25% of its nominal voltage level.



Your circuit board should include an external circuit to control the VQPS power supply. Refer to AN 057: Controlling VQPS with the Efinity SVF Player (https://www.efinixinc.com/support/docsdl.php?s=ef&pn=AN057) for details.

Recommended Energy Storage Capacitor

You should place at least one 47 uF capacitor at the output of each power regulator as energy storage.

Recommended Filtering Circuit

- π filter (C-L-C) or τ filter (L-C) of an inductor or a magnetic bead is required for every power outlet.
- The nominal current should be at least 30% higher than the current of a particular power trail.
- VCCA pins must have separate filtering circuits.

- Efinix recommends that VCC pins have separate filtering circuits.

Unused Resources and Features

For unused resources and features, follow these connection requirements:

Unused Resource/Feature	Pin	Note
PLL	VCCA_xx	Connect to VCC.
HSIO Bank	VCCIO_xx	Connect to either 1.2 V, 1.5 V, or 1.8 V.
HVIO Bank	VCCIO33_xx	Connect to either 1.8 V, 2.5 V, 3.0 V, or 3.3 V.

Power Down Sequence

Important: You must follow the following power-down sequence when powering Titanium FPGAs.

There is no specific power-down sequence. However, the VQPS power supply must follow the Fuse Programming Requirements.

Configuration Pin Checklist

Check	Pin Name	Package Pins	Description
☐	All	All	The system follows the guidelines for the SPI flash capacity. See guideline.
☐	All	All	If using internal reconfiguration, the system follows the recommended guidelines. See guideline.
☐	CBSEL[1:0]	E3 F3	The system follows the guidelines for CBSEL. See guideline.
☐	CCK	N1	Connect according to the configuration mode. See SPI Active Parallel (x4) guideline.
☐	CDI[n:0]	H4 H5 J10 J12 J13 K10 K11 K12 K13 K15 K8 K9 L1 L13 L15 L7 M1 M10 M14 M15 M6 M7 N10 N6 P10 P7 P9 R10 R11 R12 R7 R9	Connect according to the configuration mode. See SPI Active Parallel (x4) guideline.
☐	CDONE	L3	The system does not hold CDONE low (if held low, the FPGA does not enter user mode). See SPI Active Parallel (x4) guideline.
☐	CRESET_N	J5	The system pulses CRESET_N low for a minimum pulse width of 320 ns (t_{CRESET_N}) during power up or reconfiguration. See SPI Active Parallel (x4) guideline.
☐	CSI	N2	Connect according to the configuration mode. See SPI Active Parallel (x4) guideline.
☐	EXT_CONFIG_CLK	J2	In active mode, EXT_CONFIG_CLK is connected from an external clock, to be used as a configuration clock.
☐	NSTATUS	E1	(Optional) Connect NSTATUS to an LED to indicate configuration completes without errors. See guideline.
☐	SSL_N	P1	Connect according to the configuration mode. See SPI Active Parallel (x4) guideline.
☐	SSU_N	L6	Active-low chip select to the upper flash device in active x8 configuration mode (dual quad mode). Connect to weak pull-up resistors. See SPI Active Parallel (x4) guideline.
☐	TCK	N4	Pull up to VCCIO with a 10 KΩ resistor. See guideline.
☐	TDI	P3	Pull up to VCCIO with a 10 KOhm resistor. See guideline.
☐	TDO	M4	Pull up to VCCIO with a 10 KOhm resistor. See guideline.

O	TEST_N	E2	Pull up to VCCIO with a 10 KOhm resistor. See SPI Active Parallel (x4) guideline.
O	TMS	N3	Pull up to VCCIO with a 10 KOhm resistor. See guideline.

Configuration Pin General Guidelines

Selecting the Right SPI Flash Device

Titanium FPGAs support an SPI flash memory interface for active mode configuration. Use these guidelines to help choose the correct flash device for your Titanium FPGA.

- **Configuration Bits**—Ensure that your chosen flash device has enough bits to store the configuration bitstream.
  — Single image—13,960,000 bits (maximum single image bits)
  — Multiple images—Multiply 13,960,000 bits (maximum single image bits) with the number of images to determine the total bits required to store the full bitstream.
  — Daisy chain—Use the formula (i x b) + (2048 x (i - 1)) where i is the number of images and b is the configuration bits for each image. For example, a daisy chain of three FPGAs uses (3 x 13,960,000) + (2,048 x (3 - 1)) = 41,884,096 bits.
- **Configuration Bus Width**—Determine the supported configuration bus width for the SPI flash device in the following table.

Configuration Mode	Width	Availablity
Active	X1	✓
	X2	✓
	X4	✓
	X8	✓

- **SPI Clock Frequency**—Ensure that your SPI flash device supports a clock frequency that is higher than the SPI active configuration clock frequency as described in the following table.

Clock Source	Divider	Frequency (MHz) (Typical)	Sample On (Default)
Internal Oscillator	DIV8	10	Positive edge
Internal Oscillator	DIV4	20	Positive edge
Internal Oscillator	DIV2	40	Negative edge
Internal Oscillator	DIV1	80	Negative edge
External Clock	-	<40	Positive edge
		40 - 125	Negative edge

- **Required Voltage**—Make sure the voltage your SPI flash device requires is the same as the FPGA I/O bank voltage.
- **Temperature Range**—Check that the SPI flash device's temperature range is compatible with the operating temperature as described in the FPGA data sheet.

Internal Reconfiguration

`SSL_N` and `TEST_N` are dual-purpose configuration pins that can be used for internal reconfiguration. When internal reconfiguration is triggered, all GPIOs that are configured as outputs are driven high or low depending on the design. The rising time and falling time of the external network must fall within 800 ns to ensure that the configuration pins are stable and the reconfiguration is successful.

For more information about this feature, refer to AN 010: Using the Internal Reconfiguration Feature to Remotely Update Trion and Titanium FPGAs (https://www.efinixinc.com/support/docsdl.php?s=ef&pn=AN010)

Guideline for Multi-Image Select Pins

`CBSEL[1:0]` are optional multi-image selection input pins that are used with multi-image configuration mode.
If you **are** using multi-image configuration, connect `CBSEL` for the image you want to use.

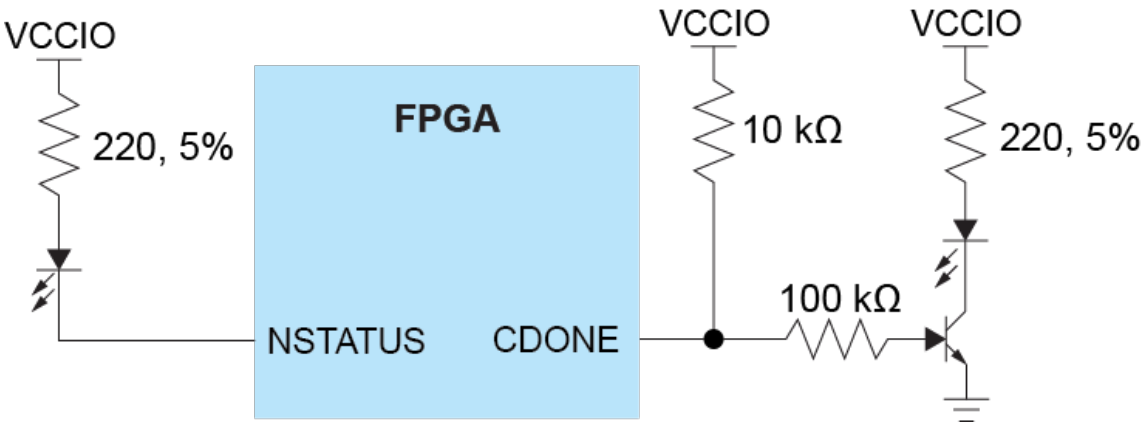
- Set `CBSEL[1:0]` to 00 for image 1
- Set `CBSEL[1:0]` to 01 for image 2
- Set `CBSEL[1:0]` to 10 for image 3
- Set `CBSEL[1:0]` to 11 for image 4

The CBSEL[1:0] pins do not require any weak pull-up or pull-down resistors if you are **not** using multi-image configuration.

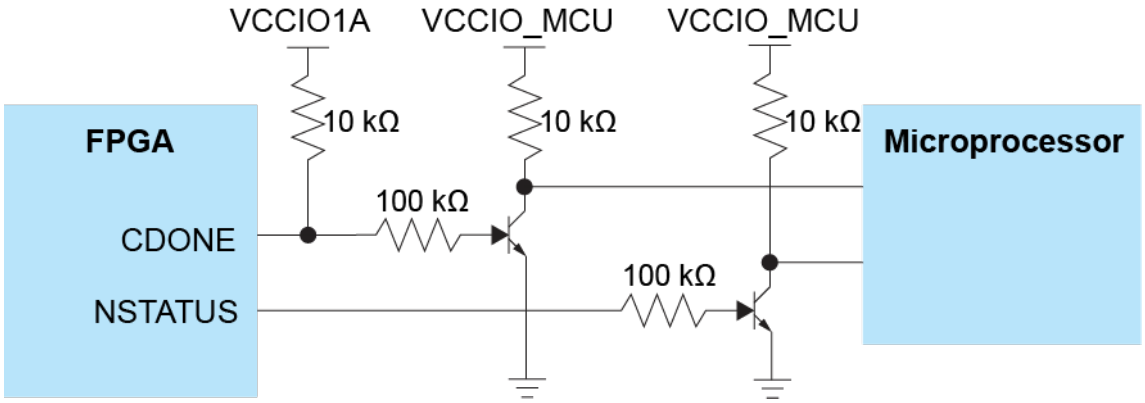
CDONE

CDONE is an open drain output; connect it to an external pull-up resistor to VCCIO. When CDONE = 1, configuration is complete. If you hold CDONE low, the device will not enter user mode.

(Optional) Connect CDONE to an LED to indicate when configuration is complete.

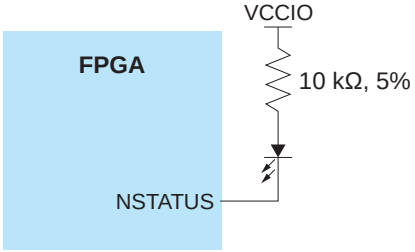


(Optional) Monitor CDONE with a microcontroller or microprocessor.



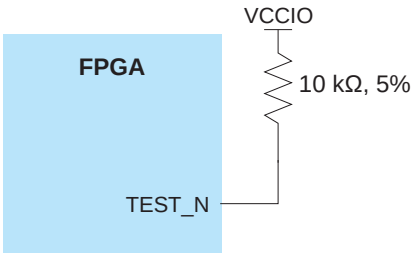
NSTATUS

(Optional) Connect NSTATUS to an LED to indicate configuration completes without errors:

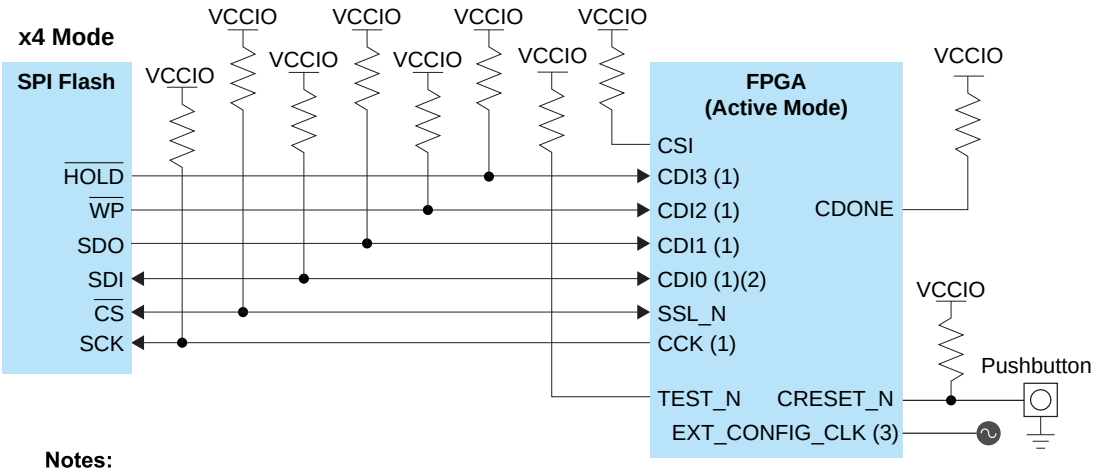


TEST_N Guidelines

During configuration, rely on the external weak pull-up or drive this pin high.



SPI Active Parallel (x4) Configuration Pin Circuitry



Notes:

1. The external pull-up is optional unless required by an external load.
2. In x4, the CDI0 pin is a bidirectional data I/O pin.
3. Connect the EXT_CONFIG_CLK to an external clock source ONLY if you have also selected **Clock Source** as the **External Clock** option in Efinity Project Editor's **Bitstream Generation** tab.

Efinix recommends that you use 10 kΩ for all unspecified pull-up and pull-down resistors in configuration circuitries. However, because of different board setups and environment noises, the configuration may fail when using 10 kΩ resistors. You can decrease the resistor values to improve signal integrity. Typically, you can also use 1 kΩ resistors.

Alternatively, you can calculate your own pull-up or pull-down resistance, R_{USER}, shown in the following sections.

Notice: The internal weak pull-up resistance, internal weak pull-down resistance, and Schmitt Trigger thresholds values used in the following formulas are included in the FPGA data sheets in the Support Center (https://www.efinixinc.com/support/docs.php).

User-Defined Pull-Up Resistor Values

$$R_{USER} = (R_{CPU} * R_{IPU}) / (R_{IPU} - R_{CPU})$$

where:

- R_{USER} = User-defined pull-up resistance
- R_{CPU} = Combined pull-up resistance
- R_{IPU} = Internal weak pull-up resistance

The combined pull-up resistance, R_{CPU}, can be derived using the following formula:

$$VT+ \leq VCCIO * (R_{IPD} / (R_{CPU} + R_{IPD}))$$

where:

- VT+ = Schmitt Trigger low-to-high threshold
- VCCIO = I/O bank power supply
- R_{IPD} = Internal weak pull-down resistance

User-Defined Pull-Down Resistor Values

$R_{USER} = (R_{CPD} * R_{IPD}) / (R_{IPD} - R_{CPD})$

where:

- R_{USER} = User-defined pull-down resistance
- R_{CPD} = Combined pull-down resistance
- R_{IPD} = Internal weak pull-down resistance

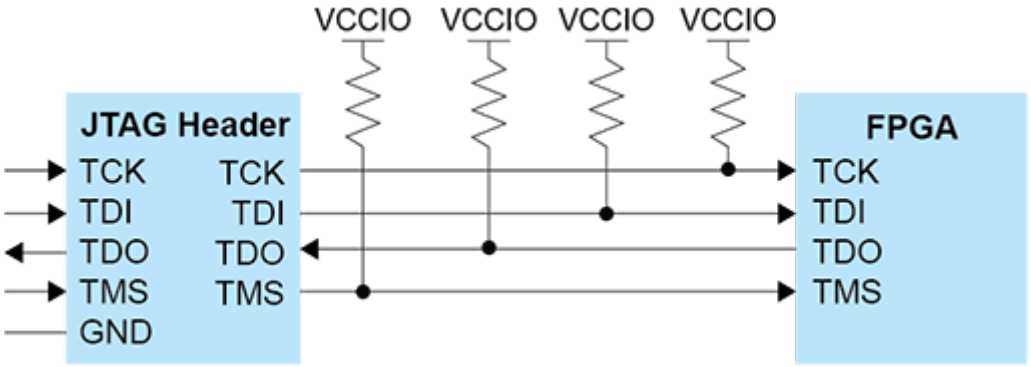
The combined pull-down resistance, R_{CPD}, can be derived using the following formula:

$VT- \geq VCCIO * (R_{CPD} / (R_{CPD} + R_{IPU}))$

where:

- VT- = Schmitt Trigger high-to-low threshold
- VCCIO = I/O bank power supply
- R_{IPU} = Internal weak pull-up resistance

JTAG Configuration Pin Circuitry



HVIO Pin Checklist

Check	Pin Name	Package Pins	Description
☐	All	All	Efinix recommends that you limit the number of 3.0 or 3.3 V HVIO as I/O or output to 6 per bank to avoid switching noise. The Efinity software issues a warning if you exceed the recommended limit.

HVIO Pin General Guidelines

Unused HVIO Banks (VCCIO)

If you are not using any of the HVIO pins in a bank, connect the bank's VCCIO33 pins to 1.8 V, 2.5 V, 3.0 V, or 3.3 V. The VCCIOxx pins for unused HVIO banks do not require decoupling capacitors.

Unused HVIO Pins

Leave unused HVIO pins floating.

HSIO Pin Checklist

Check	Pin Name	Package Pins	Description
☐	All	All	The system follows the guidelines for differential traces. See guideline.
☐	All	All	The system follows the guidelines for impedance matching. See guideline.
☐	All	All	The system follows the guidelines for avoiding crosstalk. See guideline.

☐	All	All	The system follows the guidelines for HSIO pins used as GPIO. See guideline.
☐	CLKn	A8 B8 C8 C9 D8 E8 G1 G14 G2 H1 H10 H11 H12 H13 H14 H15 H2 H3 J1 J15 J2 K1 K7 L8 L9 M8 M9 N8 P8 R8	☐ My design uses these global clock network inputs as dedicated clock input pins for input clocks. ☐ My system uses these global control network input pins for high fanout and global reset.
☐	GPIOT*, GPIOB*, GPIOR*, GPIOL*	A10 A11 A12 A13 A14 A2 A3 A4 A5 A6 A7 A8 A9 B1 B10 B11 B12 B14 B2 B3 B5 B6 B7 B8 B9 C1 C10 C12 C13 C14 C15 C2 C3 C4 C5 C6 C8 C9 D1 D10 D12 D13 D14 D15 D5 D7 D8 E1 E10 E12 E14 E15 E2 E3 E6 E7 E8 E9 F1 F10 F11 F12 F13 F15 F2 F3 F5 F7 F8 F9 G1 G11 G13 G14 G15 G2 G3 G4 G5 H1 H10 H11 H12 H13 H14 H15 H2 H3 H4 H5 J1 J10 J12 J13 J14 J15 J2 J3 K1 K10 K11 K12 K13 K14 K15 K2 K3 K4 K6 K7 K8 K9 L1 L11 L12 L13 L15 L6 L7 L8 L9 M1 M10 M14 M15 M2 M6 M7 M8 M9 N1 N10 N13 N14 N2 N6 N8 P1 P10 P11 P12 P13 P14 P15 P2 P5 P6 P7 P8 P9 R10 R11 R12 R13 R14 R2 R3 R4 R5 R6 R7 R8 R9	☐ Titanium FPGAs have a 100 Ohm on-die termination resistor for these pins. Therefore, do not use external termination. ☐ When used as GPIO, these HSIO pins follow the guidelines. See guidelines.
☐	REF_RES	B15 D11 E4 F6 K5 M12 M5 N12	The system follows the guidelines for REF_RES. See guideline.
☐	VREF	C10 E1 E6 G15 H5 K7 L15 L6	☐ When configuring an I/O pad of the standard's input path as a VREF pin, you must use the VREF from the same physical I/O bank even when the I/O banks are merged to share a common VCCIO pin. ☐ When using an HSIO with the HSTL or SSTL I/O standards, you must configure an I/O pad of the standard's input path as a VREF pin

HSIO Pin General Guidelines

Differential Traces

Follow these design guidelines for HSIO traces:

- The P and N channel in an HSIO pair should be as close to each other as possible, and the distance between the pair's P and N channel should be constant.
- The HSIO TX and RX should be as close as possible to any connectors.
- To minimize skew, the electrical length of traces:
  - between the differential HSIO P and N channels should be +/- 0.15 mm
  - between one HSIO pair to another HSIO pair should be +/- 1.50 mm
- Minimize the number of vias on the signal path.
- If there is any parasitic loading (such as capacitance), it should be in equal amounts to each line.
- Traces should be bend >135 degrees to avoid signal loss and jitter.

Impedance Matching

The HSIO trace impedance should be:

- HSIO differential: 100 ohms +/- 10%
- HSIO single-ended: 50 ohms +/- 10%

Avoiding Crosstalk between HSIO and Single-Ended Signals

Differential HSIO signals must be isolated from single-ended signals; the single-ended signals may cause interference on the differential signals. You can implement the isolation by:

- Designing the trace for differential HSIO pairs to be at least 12 mm from the trace for single-ended I/O, if both are on the same PCB

layer.
- Isolating the differential HSIO pair signals from the single-ended signals using power or ground planes.
- Minimize stub lengths because they increase reflection noise.

HSIO Used as GPIO

You can use the HSIO pins as 1.2 V, 1.5 V, or 1.8 V single-ended GPIO pins.

When you are using an HSIO pin as a GPIO, make sure to leave at least 1 pair of unassigned HSIO pins between any GPIO and LVDS or MIPI lane pins. This rule applies for pins on each side of the device (top, bottom, left, right). This separation reduces noise. The Efinity software issues an error if you do not leave this separation.

Separate the I/O placement evenly in a single bank.

Unused HSIO Banks or Pins

If you are not using any of the HSIO in a bank, connect the bank's `VCCIOxx` pins to 1.2 V, 1.5 V, or 1.8 V.

The `VCCIOxx` pins for unused HSIO banks do not require decoupling capacitors.

Leave unused HSIO pins floating.

Termination

On-board termination is not required.

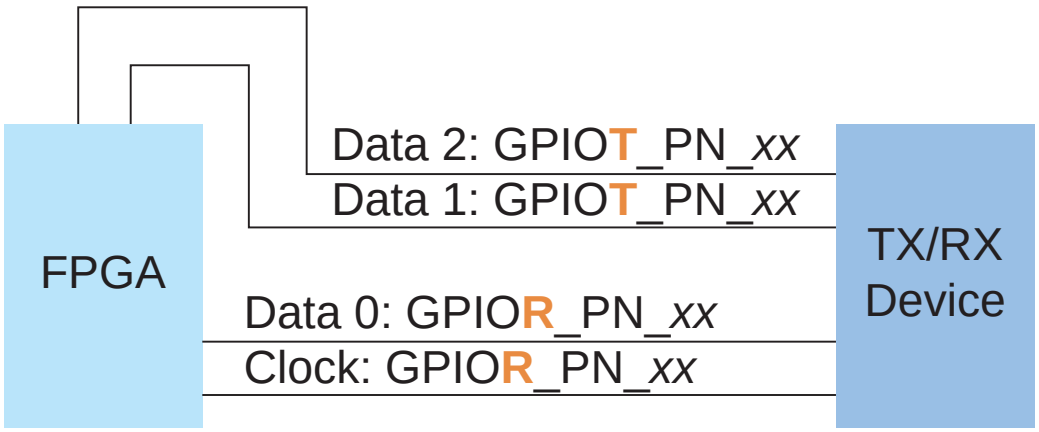
Differential Lanes in the Same Channel Should Use GPIOs on the Same Side

An LVDS differential channel that contains a clock lane and data lanes to the same TX/RX destination must use GPIO pins on same side of the FPGA.

Example: An LVDS channel with a clock lane using GPIO**R**_PN_xx must use data lanes on the right side (GPIO**R**_PN_xx).

Incorrect

Data lanes are not on the same side as the clock lane



Correct

Data and clock lanes are on the same side



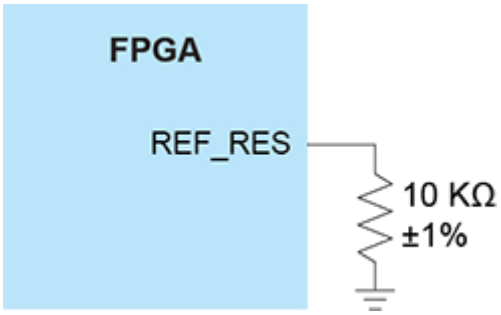
REF_RES Pin

REF_RES is a reference resistor to generate constant current for the related circuits.

Connect the following REF_RES pins to ground through a 10 kOhm resistor with a tolerance of +/-1%:

- REF_RES_2A and REF_RES_4A pins must be connected.
- REF_RES pin of the particular bank, if pins in the bank are used as LVDS TX or MIPI TX lane.
- REF_RES_3A pin, if the internal oscillator is used.
- REF_RES_3A pin, if blowing of fuses for FPGA security is required.

You can leave the REF_RES pins floating if none of the above are applicable.



MIPI Lane Pin Checklist

Check	Pin Name	Package Pins	Description
☐	All	All	Refer to the Titanium MIPI Interface PCB Design User Guide (https://www.efinixinc.com/support/docsdl.php?s=ef&pn=PCB-MIPI-TI) for detailed guidelines.
☐	RX_CLK*	A10 A5 A6 B10 C1 C13 D1 D13 D8 E8 G1 H1 H12 H13 K7 L8 M14 M15 M2 M6 N2 N6 P10 R10	The system follows the guidelines for MIPI traces. See guidelines.
☐	RX_DATA*	A11 A2 A7 A8 A9 B1 B11 B2 B6 B7 B8 B9 C10 C14 C15 C2 C6 C8 C9 D10 D14 D15 D5 D7 E1 E10 E12 E14 E15 E2 E3 E6 E7 E9 F1 F10 F11 F12 F13 F15 F2 F3 F5 F7 F8 F9 G11 G13 G14 G15 G2 G3 G4 G5 H10 H11 H14 H15 H2 H3 H4 H5 J1 J10 J12 J13 J14 J15 J2 J3 K1 K10 K11 K12 K13 K14 K15 K2 K3 K4 K6 K8 K9 L1 L11 L12 L13 L15 L6 L7 L9 M1 M10 M7 M8 M9 N1 N10 N8 P1 P11 P12 P2 P5 P6 P7 P8 P9 R11 R12 R2 R5 R6 R7 R8 R9	☐ The system follows the guidelines for MIPI traces. See guidelines. ☐ When using HSIO pins as GPIO, make sure to leave at least 1 pair of unassigned HSIO pins between any GPIO and HSIO used as MIPI lanes. This rule applies for pins on each side of the device (top, bottom, left, right). This separation reduces noise. The Efinity software issues an error if you do not leave this separation.

MIPI Lane Pin General Guidelines

MIPI Traces

To reduce signal reflections and impedance changes, route high-speed signals using:

- A minimum number of vias and corners
- The same PCB layer for all the lanes in the same channel

The MIPI trace impedance should be:
Data differential 100 Ohm +/- 10%
Data single-ended 50 Ohm +/- 10%

Match routing length for 1.5 Gbps:
Data P/N pair +/- 0.10 mm
Clock +/- 1.00 mm

Match routing length for 1.0 Gbps:
Data P/N pair +/- 0.15 mm
ClocK +/- 1.5 mm

Unused MIPI Lanes

You can leave unused MIPI lanes floating.

PLL Pin Checklist

Check	Pin Name	Package Pins	Description
☐	All	All	The PLL reference clock must be between 16 and 800 MHz.
☐	PLLINn	A11 B2 C14 C5 E6 L11 P11 P2 R13 R5	Assign the PLL reference clock port based on the PLL that you are using. See guideline.

PLL Pin General Guidelines

Unused PLLs

Always connect a VCCA_PLL pin to VCC regardless of whether you use the PLL or not.

The VCCA_PLL pins for unused PLLs do not require decoupling capacitors.

PLL Reference Clocks

The PLL reference clock port can be driven by single ended I/O pins or a differential pair clock.

PLL Reference Clock Resource Assignments

PLL	REFCLK0	REFCLK1	REFCLK2	External Feedback I/O
PLL_BL	Single-ended: GPIOL_P_00_PLLIN0 Differential: GPIOL_P_00_PLLIN0, GPIOL_N_00	Single-ended: GPIOB_P_00_PLLIN1 Differential: GPIOB_P_00_PLLIN1, GPIOB_N_00	Unbonded (1)	Single-ended: GPIOB_P_01_EXTFB Differential: GPIOB_P_01_EXTFB, GPIOB_N_01
PLL_TL	Single-ended: GPIOL_P_18_PLLIN0 Differential: GPIOL_P_18_PLLIN0, GPIOL_N_18	Single-ended: GPIOT_P_00_PLLIN1 Differential: GPIOT_P_00_PLLIN1 GPIOT_N_00	GPIOL_11_PLLIN2	Single-ended: GPIOL_P_17_EXTFB Differential: GPIOL_P_17_EXTFB, GPIOL_N_17
PLL_TR	Single-ended: GPIOR_P_19_PLLIN0 Differential: GPIOR_P_19_PLLIN0, GPIOR_N_19	Single-ended: GPIOT_P_17_PLLIN1 Differential: GPIOT_P_17_PLLIN1, GPIOT_N_17	Unbonded (1)	Single-ended: GPIOT_P_16_EXTFB Differential: GPIOT_P_16_EXTFB, GPIOT_N_16
PLL_BR	Single-ended: GPIOR_P_00_PLLIN0 Differential: GPIOR_P_00_PLLIN0, GPIOR_N_00_CDI22	Single-ended: GPIOB_P_17_PLLIN1 Differential: GPIOB_P_17_PLLIN1, GPIOB_N_17	GPIOR_29_PLLIN2	Single-ended: GPIOR_P_01_EXTFB Differential: GPIOR_P_01_EXTFB, GPIOR_N_01_CDI23

1. There is no dedicated pin assigned to this reference clock.

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